2.19 Trigonometry: Small Angle Approximations

Worked Solutions 1–24

1)

$$\sin\theta = \frac{h}{r} \Rightarrow h = r\sin\theta \text{ and } \tan\theta = \frac{h}{y} \Rightarrow h = y\tan\theta \text{ so } r\sin\theta = y\tan\theta$$

As $\theta \to 0$, $h \to r\theta$ and $y \to r$. So $r\sin\theta \approx r\tan\theta \approx r\theta \Rightarrow \sin\theta \approx \tan\theta \approx \theta$

2)

$\cos 2x = 1 - 2\sin^2 x$

Let $x = \frac{\theta}{2}$ so $\cos\theta = 1 - 2\sin^2(\frac{\theta}{2}) \approx 1 - 2(\frac{\theta}{2})^2$. So $\cos\theta \approx 1 - \frac{\theta^2}{2}$

3 a)

$\theta \times \theta = \theta^2$

3 b)

$1 - \frac{\theta^2}{2} - 1 = -\frac{\theta^2}{2}$

3 c)

$1 - \frac{(6\theta)^2}{2} = 1 - 18\theta^2$

3 d)

$\theta - 2\theta = -\theta$

3 e)

$\theta \times 3\theta + 1 - \frac{(3\theta)^2}{2} = 3\theta^2 + 1 - \frac{9}{2}\theta^2 = 1 - \frac{3}{2}\theta^2$

4 a)

$\frac{\theta^2}{1-1-\frac{\theta^2}{2}} = \frac{\theta^2}{\frac{\theta^2}{2}} = 2$

4 b)

$(1 - \frac{\theta^2}{2})(1 - \frac{(3\theta)^2}{2}) = 1 - \frac{\theta^2}{2} - \frac{9\theta^2}{2} + \frac{9\theta^4}{4} = 1 - 5\theta^2$

4 c)

$(1 - \frac{\theta^2}{2})(1 - \frac{\theta^2}{2}) \approx 1 - \theta^2$

4 d)

$\theta^2 + 1 - \theta^2 = 1$

4 e)

$\frac{\theta^3}{1-(1-\frac{(2\theta)^2}{2})} = \frac{\theta^3}{2\theta^2} = \frac{1}{2}\theta$

4 f)

$$\frac{1-\frac{(8\theta)^2}{2}-(1-\frac{(4\theta)^2}{2})}{8\theta-4\theta}=\frac{1-32\theta^2-1+8\theta^2}{4\theta}=\frac{-24\theta^2}{4\theta}=-6\theta$$

4 g)

$\frac{1-(1-\frac{(2\theta)^2}{2})}{6\theta}=\frac{2\theta^2}{6\theta}=\frac{1}{3}\theta$

5 a)

$\cos 4\theta \approx 1-\frac{(4\theta)^2}{2}\approx 1-8\theta^2$

So $\cos 0.4 \approx 1-8(0.1)^2 \approx 1-0.08 \approx 0.92$. $\cos 0.4 = 0.9210610\ldots$

5 b)

Percentage error is $\frac{0.0010610\ldots}{0.9210610\ldots}\times 100 = 0.115\%$

6 a)

$\frac{\theta}{\sin\theta}\approx\frac{\theta}{\theta}\approx 1$

6 b)

$\frac{1}{\tan 2\theta}\approx\frac{1}{2\theta}$

6 c)

$\frac{1}{\tan 4\theta}-\frac{1}{\sin 2\theta}\approx\frac{1}{4\theta}-\frac{1}{2\theta}=-\frac{1}{4\theta}$

7 a)

$\frac{\sin x+\cos x}{\cos x}=\tan x+1\approx x+1$

7 b)

$\frac{x+1}{1}\approx x+1$

7 c)

$$\frac{\sin 0.1+\cos 0.1}{\cos 0.1}\approx 1.1 \text{ and } \frac{\sin 0.1+\cos 0.1}{\cos 0.1}=1.100335\ldots$$

Percentage error is $\frac{0.000335\ldots}{1.100335\ldots}\times 100 = 0.0304\%$

8 a)

$-2+1-\frac{\theta^2}{2}+\theta^2\approx -1+\frac{1}{2}\theta^2$ as req.

8 b)

$$-1+\tfrac{1}{2}\theta^2\approx 4\theta$$

$\frac{1}{2}\theta^2-4\theta-1=0$ has solutions $4\pm 3\sqrt{2}$. $\theta\approx -0.243$ or $\theta\approx 8.24$. But θ is small so $\theta\approx -0.243$.

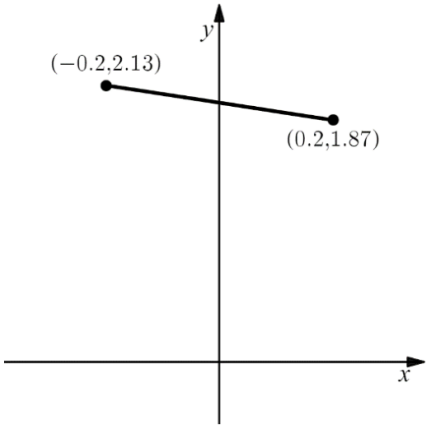
9)

$$\frac{1-(1-\frac{(0.1x)^2}{2})}{x} \approx \frac{x^2/200}{x} \approx \frac{1}{200}x \text{ and } \frac{1}{5}-x$$
$$\frac{1}{200}x \approx \frac{1}{5}-x \text{ so } x \approx \frac{40}{201} \approx 0.199$$

10 a)

$\mathrm{f}(x) \approx 2+4\times\frac{x}{2}-8\times\frac{x}{3} \approx 2-\frac{2}{3}x$

10 b)



11)

When x is small $\cos x \approx 1-\frac{x^2}{2}$

$$2x-\tfrac{1}{4}+1-\tfrac{x^2}{2}=0 \Rightarrow 2x+\tfrac{3}{4}-\tfrac{x^2}{2}=0 \Rightarrow \tfrac{x^2}{2}-2x-\tfrac{3}{4}=0$$

$$x=2\pm\sqrt{\tfrac{11}{2}} \text{ so } x\approx -0.345 \text{ or } x\approx 4.35$$

Since x is small $x \approx -0.345$

12)

$$x+x(1-\tfrac{(4x)^2}{2}) \approx x+x(1-8x^2) \approx x+x-8x^3 \approx 2x$$

$$\int_0^{0.3}(2x)^{1/3}\,dx = \int_0^{0.3}2^{1/3}x^{1/3}\,dx = [2^{1/3}\times\tfrac{3}{4}x^{4/3}]_0^{0.3}$$
$$= (0.60249\ldots)-(0) = 0.189772\ldots$$

Which is approximately 0.190 as req.

13 a)

$1-\frac{\theta^2}{2}-(1-\frac{(2\theta)^2}{2}) = 1-\frac{\theta^2}{2}-1+\frac{4\theta^2}{2} = -\frac{\theta^2}{2}+2\theta^2 = \frac{3}{2}\theta^2$

13 b)

$\lim_{\theta\to 0}\frac{\cos\theta-\cos 2\theta}{\theta^2}=\lim_{\theta\to 0}\frac{\frac{3}{2}\theta^2}{\theta^2}=\lim_{\theta\to 0}\frac{3}{2}=\frac{3}{2}$

14 a)

$$\lim_{h\to 0}\frac{\cos h-1}{h}=\lim_{h\to 0}\frac{1-\frac{h^2}{2}-1}{h}=\lim_{h\to 0}\frac{-\frac{h^2}{2}}{h}=\lim_{h\to 0}\frac{h}{2}=0$$

14 b)

$\lim_{h\to 0}\frac{\sin h}{h}=\lim_{h\to 0}\frac{h}{h}=\lim_{h\to 0}1=1$

15 a)

$\mathrm{f}(\theta)=\sin\theta$

$$\begin{aligned}\frac{dy}{d\theta}&=\lim_{h\to 0}\frac{\sin(\theta+h)-\sin\theta}{h}=\lim_{h\to 0}\frac{\sin\theta\cos h+\sin h\cos\theta-\sin\theta}{h}\\&=\lim_{h\to 0}\frac{\sin\theta(\cos h-1)+\cos\theta(\sin h)}{h}\\&=\lim_{h\to 0}\frac{\sin\theta(1-\frac{h^2}{2}-1)+\cos\theta(h)}{h}\\&=\lim_{h\to 0}\frac{-\frac{1}{2}h^2\sin\theta+h\cos\theta}{h}=\lim_{h\to 0}(-\tfrac{1}{2}h\sin\theta+\cos\theta)=\cos\theta\end{aligned}$$

15 b)

$\mathrm{f}(\theta)=\cos\theta$

$$\begin{aligned}\frac{dy}{d\theta}&=\lim_{h\to 0}\frac{\cos(\theta+h)-\cos\theta}{h}=\lim_{h\to 0}\frac{\cos\theta\cos h-\sin h\sin\theta-\cos\theta}{h}\\&=\lim_{h\to 0}\frac{\cos\theta(\cos h-1)-\sin\theta(\sin h)}{h}\\&=\lim_{h\to 0}\frac{\cos\theta(1-\frac{h^2}{2}-1)-\sin\theta(h)}{h}\\&=\lim_{h\to 0}\frac{-\frac{1}{2}h^2\cos\theta-h\sin\theta}{h}=\lim_{h\to 0}(-\tfrac{1}{2}h\cos\theta-\sin\theta)=-\sin\theta\end{aligned}$$

16)

$$\mathrm{f}(\tfrac{\pi}{3})=\cos\tfrac{\pi}{3}=\tfrac{1}{2},\quad \mathrm{f}(\tfrac{\pi}{3}+h)=\cos(\tfrac{\pi}{3}+h)$$

$$\begin{aligned}\frac{dy}{d\theta}&=\lim_{h\to 0}\frac{\cos(\frac{\pi}{3}+h)-\frac{1}{2}}{h}\\&=\lim_{h\to 0}\frac{\cos\frac{\pi}{3}\cos h-\sin\frac{\pi}{3}\sin h-\frac{1}{2}}{h}\\&=\lim_{h\to 0}\frac{\frac{1}{2}\cos h-\frac{\sqrt{3}}{2}\sin h-\frac{1}{2}}{h}\\&=\lim_{h\to 0}\frac{\frac{1}{2}(\cos h-1)-\frac{\sqrt{3}}{2}\sin h}{h}\end{aligned}$$

$$= \lim_{h\to 0} \frac{\frac{1}{2}(1-\frac{h^2}{2}-1)-\frac{\sqrt{3}}{2}h}{h}$$
$$= \lim_{h\to 0} \frac{-\frac{1}{4}h^2-\frac{\sqrt{3}}{2}h}{h} = \lim_{h\to 0}(-\tfrac{1}{4}h-\tfrac{\sqrt{3}}{2}) = -\tfrac{\sqrt{3}}{2}$$

17)

$\mathrm{f}(\theta) = \tan\theta$

$$\frac{dy}{d\theta} = \lim_{h\to 0} \frac{\tan(\theta+h)-\tan\theta}{h}$$
$$= \lim_{h\to 0} \frac{\frac{\sin(\theta+h)}{\cos(\theta+h)} - \frac{\sin\theta}{\cos\theta}}{h}$$
$$= \lim_{h\to 0} \frac{\frac{\sin(\theta+h)\cos\theta - \cos(\theta+h)\sin\theta}{\cos\theta\cos(\theta+h)}}{h}$$
$$= \lim_{h\to 0} \frac{\sin(\theta+h)\cos\theta-\cos(\theta+h)\sin\theta}{h\cos\theta\cos(\theta+h)}$$

Using the compound angle formula:

$$= \lim_{h\to 0} \frac{\sin(\theta+h-\theta)}{h\cos\theta\cos(\theta+h)} = \lim_{h\to 0} \frac{\sin h}{h\cos\theta\cos(\theta+h)}$$

For small angles:

$$= \lim_{h\to 0} \frac{h}{h\cos\theta\cos(\theta+h)} = \lim_{h\to 0} \frac{1}{\cos\theta\cos(\theta+h)} = \frac{1}{\cos^2\theta} = \sec^2\theta$$

18)

$\mathrm{f}(\theta) = \sec\theta$

$$\frac{dy}{d\theta} = \lim_{h\to 0} \frac{\frac{1}{\cos(\theta+h)} - \frac{1}{\cos\theta}}{h}$$
$$= \lim_{h\to 0} \frac{1}{h}\left[\frac{1}{\cos(\theta+h)} - \frac{1}{\cos\theta}\right]$$
$$= \lim_{h\to 0} \frac{1}{h}\left[\frac{\cos\theta-\cos(\theta+h)}{\cos(\theta+h)\cos\theta}\right]$$

Using the compound angle formula:

$$= \lim_{h\to 0} \frac{1}{h}\left[\frac{\cos\theta-\cos\theta\cos h+\sin\theta\sin h}{\cos(\theta+h)\cos h}\right]$$
$$= \lim_{h\to 0} \frac{1}{h}\left[\frac{\cos\theta(1-\cos h)+\sin\theta\sin h}{\cos(\theta+h)\cos\theta}\right]$$

For small angles:

$$= \lim_{h\to 0} \frac{1}{h}\left[\frac{\cos\theta(1-1+\frac{1}{2}h^2)+h\sin\theta}{\cos(\theta+h)\cos\theta}\right]$$
$$= \lim_{h\to 0} \frac{\frac{1}{2}h\cos\theta+\sin\theta}{\cos(\theta+h)\cos\theta}$$
$$= \frac{\sin\theta}{\cos^2\theta} = \sec\theta\tan\theta \text{ as req.}$$

19 a)

$$(1-8x)^{1/2} \approx 1 + \tfrac{1}{2}(-8x) + \tfrac{1}{2}(\tfrac{1}{2})(-\tfrac{1}{2})(-8x)^2 \approx 1 - 4x - 8x^2$$

19 b)

$$\sin 2x + \sqrt{\cos 4x} \approx 2x + (1 - \tfrac{(4x)^2}{2})^{1/2} \approx 2x + (1-8x^2)^{1/2}$$

Sub x^2 into result from before: $1 - 4x^2 - 8x^4$. Therefore $\approx 2x + (1-8x^2)^{1/2} \approx 2x + 1 - 4x^2 - 8x^4$. Ignoring x^4 terms $\approx 1 + 2x - 4x^2$.

20 a)

$$(1 - \tfrac{1}{2}x^2)^{1/3} \approx 1 + \tfrac{1}{3}(-\tfrac{1}{2}x^2) \approx 1 - \tfrac{1}{6}x^2$$

20 b)

$$\begin{aligned}\int_0^{0.2} (\cos x)^{1/3}\,dx &\approx \int_0^{0.2} (1 - \tfrac{1}{2}x^2)^{1/3}\,dx \\ &\approx \int_0^{0.2} (1 - \tfrac{1}{6}x^2)\,dx \approx [x - \tfrac{1}{18}x^3]_0^{0.2} \\ &\approx (0.199556) - (0) = 0.1996 \text{ (4 dp)}\end{aligned}$$

21 a)

$$\frac{dx}{dy} = 2\sec^2 4y = \frac{2}{\cos^2 4y} \Rightarrow \left.\frac{dx}{dy}\right|_{y=0} = 2 \text{ so } \left.\frac{dy}{dx}\right|_{(0,0)} = \tfrac{1}{2}$$

21 b)

$$\tfrac{1}{2}\tan 4y \approx \tfrac{1}{2} \times 4y \text{ so } x \approx 2y$$
$$\left.\frac{dx}{dy}\right|_{(0,0)} = 2 \Rightarrow \left.\frac{dy}{dx}\right|_{(0,0)} = \tfrac{1}{2}$$

22 a)

When $x = 0$, $16y = \frac{\pi}{2} \Rightarrow y = \frac{\pi}{32}$.

$$\frac{dx}{dy} = -48\sin 16y$$
$$\left.\frac{dx}{dy}\right|_{y=\pi/32} = -48 \text{ so } \frac{dy}{dx} = -\tfrac{1}{48}$$

22 b)

$$3\cos 16y \approx 3(1 - \tfrac{(16y)^2}{2}) \approx 3 - 384y^2$$

So $x \approx 3 - 384y^2$. When $x = 0$, $y \approx \frac{\pi}{32}$.

$$\frac{dx}{dy} = -768y \Rightarrow \left.\frac{dx}{dy}\right|_{y=\pi/32} = -24\pi \text{ so } \frac{dy}{dx} \approx -\tfrac{1}{24\pi}$$

23 a)

$\frac{1}{2} - x - 2x \approx 0 \Rightarrow 3x \approx \frac{1}{2} \Rightarrow x \approx \frac{1}{6} \approx 0.167$

23 b)

$\mathrm{f}(x) = \frac{1}{2} - x - \sin 2x$ so $\mathrm{f}'(x) = 1 - 2\cos 2x$

$$x_{n+1} = x_n - \frac{\frac{1}{2} - x_n - \sin 2x_n}{-1 - 2\cos 2x_n}$$

$x_0 = \frac{1}{6}$, $x_1 = 0.16879\ldots$, $x_2 = 0.16879$ so $x \approx 0.169$.

24 a)

$$2 + 5x + x \approx 1 - \tfrac{x^2}{2} \Rightarrow \tfrac{x^2}{2} + 6x + 1 \approx 0$$

$$x \approx -0.169 \text{ or } x \approx -11.8 \text{ but } x \text{ is small so } x \approx -0.169$$

24 b)

$\mathrm{f}(x) = 2 + 5x + \sin x - \cos x$ so $\mathrm{f}'(x) = 5 + \cos x + \sin x$

$$x_{n+1} = x_n - \frac{2 + 5x_n + \sin x_n - \cos x_n}{5 + \cos x_n + \sin x_n}$$

$x_0 = -\frac{1}{6}$, $x_1 = -0.1691798\ldots$, $x_2 = -0.169180\ldots$ so $x \approx -0.169$.